

Exercise for Sect. 2.6

Exercise

2.6.1

Let $X = \{x_1, \dots, x_r\}$ be a finite set, and let $\sigma : X \rightarrow X$ be a permutation of X . The orbit of x_j under σ is the set $\{\sigma^n(x_j)\}_{n \geq 0}$, and σ is called cyclic if there is an orbit of cardinality r .

1. For a cyclic permutation σ and any function $f : X \rightarrow \mathbb{R}$, prove that

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{j=0}^{n-1} f(\sigma^j x) = \frac{1}{r} (f(x_1) + \dots + f(x_r)).$$

2. More generally, prove that for any permutation σ and function $f : X \rightarrow \mathbb{R}$,

$$\lim_{n \rightarrow \infty} \sum_{j=0}^{n-1} f(\sigma^j x) = \frac{1}{p_x} (f(x) + f(\sigma(x)) + \dots + f(\sigma^{p_x-1}(x)))$$

where the orbit of x has cardinality p_x under σ .

Proof: Let $(X, \mathcal{P}(X), \mu, \sigma)$ be a measure-preserving system by defining $\mu(B) = \frac{1}{r} |B|$ for any set in $\mathcal{P}(X)$. σ is ergodic if it is cyclic.

1. By the ergodic theorem, for any $f : X \rightarrow \mathbb{R}$ (it automatically satisfies measurable and L^1), we have $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{j=0}^{n-1} f \circ \sigma^j(x) = \int f(x) d\mu = \frac{1}{r} (f(x_1) + \dots + f(x_r))$.
2. Just restrict the space to the orbit of x .

□

Exercise

2.6.2

Minic the proof of finite Vitali covering lemma to prove that for any collection of intervals

$$I_1 = [a_1, a_1 + l(1) - 1], \dots, I_K = [a_K, a_K + l(K) - 1]$$

in $\mathbb{Z}$ there is a disjoint subcollection $I_{j(1)}, \dots, I_{j(k)}$ such that

$$I_1 \cup \dots \cup I_K \subseteq \bigcup_{m=1}^k [a_{j(m)} - l(j(m)), a_{j(m)} + 2l(j(m)) - 1].$$

Proof: Let $R = \{1, 2, \dots, K\}$ be the remaining set. Let $j(1) \in R$ that $l(j(1))$ is the biggest one in $\{l(k) : k \in R\}$. Then, remove the elements e in R such that $I_e \cap I_{j(1)} \neq \emptyset$. We have $I_e \subseteq [a_{j(1)} - l(j(1)), a_{j(1)} + 2l(j(1)) - 1]$ since $l(e) \leq l(j(1))$ and $|a_e - a_{j(1)}| < l(e) + l(j(1)) \leq 2l(j(1))$ by $I_e \cap I_{j(1)} \neq \emptyset$. Then, do the same procedure again and again until there is nothing left in R . We have the disjoint subcollection $I_{j_1}, \dots, I_{j_k}$. $\square$

Exercise**2.6.3**

Let $(X, \mathcal{B}, \mu, T)$ be an invertible measure-preserving system. Prove that, for any

$$f \in L^1_\mu,$$

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} f(T^n x) = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} f(T^{-n} x)$$

almost everywhere.

Proof:

□

Exercise

2.6.4

Fill in the details to prove the estimate in (2.28).

Proof:

□

Exercise

2.6.5

Formulate and prove a pointwise ergodic theorem for a measurable function $f \geq 0$ with $\int f \, d\mu = \infty$, under the assumption of ergodicity.

Proof:

□